LAB 11

ML-Signal Stage: a Walk-Forward Classifier from Probability to Position

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	ML signal
TSC reference	A4, K4, K6
Duration	20 min
Folder	labs_fullcode/Lab11-ml-signal-classifier/
Files	Lab11-ml-signal-classifier/ml_stage.py, pipeline/ml.py, Lab11-ml-signal-classifier/solution/ml.py
Prompts	Lab11_Prompts.pdf in this folder (step -> prompt, ready to copy)
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

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Scenario

The desk's data scientist claims a machine-learning model reading technical features can predict whether SPY closes up tomorrow. The pipeline will only accept an ML signal that is validated walk-forward, turned into positions with an explicit probability threshold and traded next bar with the same costs as every other signal - and that still beats buy and hold.

Objective

Build pipeline/ml.py: engineer causal features, train logistic regression, random forest and XGBoost with walk-forward TimeSeriesSplit validation, turn P(up) into a target position and backtest it after costs.

Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

How to Run the Python Scripts

Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.

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Step	Command / file
Open a terminal in the labs_fullcode folder	cd labs_fullcode
Create and activate a virtual environment (once)	python3 -m venv .venv source .venv/bin/activate (Windows: .venv\Scripts\activate)
Install the libraries (once)	pip install -r requirements.txt
Optional: your Alpaca paper keys (leave blank to run OFFLINE)	cp .env.example .env (Windows: copy .env.example .env) then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env
Run Lab 11	python3 Lab11-ml-signal-classifier/ml_stage.py (same as: python -m pipeline ml)
The complete code	pipeline/ml.py (reference copy: Lab11-ml-signal-classifier/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

Lab Steps - In Python with Your AI Assistant

AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab11-ml-signal-classifier/Lab11_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.

Step 1: Build causal features and the label (K4)

Ask your assistant for build_features().

AI prompt - paste into your AI coding assistant
Create labs_fullcode/pipeline/ml.py with build_features(bars): features known at the close of day t only - ret_lag1/2/3/5 (r.shift(lag-1), so ret_lag1 = today's return), mom_5, mom_21, rsi_14 and rsi_2 (pipeline.indicators.ta_rsi), macd_hist / close (ta_macd), Bollinger %B from ta_bb, vol_21 annualised, dist_sma50 and dist_sma200, volume z-score over 21 days; label target = 1 if close t+1 > close t. Drop the last row (no label yet) and rows with NaN. The label may use shift(-1); no FEATURE may. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: build_features(load_bars('SPY','2018-01-01','2025-12-31')) has 1,811 rows and 15 columns (14 features + target).

Step 2: Add walk-forward validation (K4, K6)

Ask your assistant for the models and the walk-forward loop.

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/ml.py add MODELS = {'Logistic regression': StandardScaler + LogisticRegression(C=0.1, max_iter=1000), 'Random forest': RandomForestClassifier(n_estimators=300, max_depth=4, min_samples_leaf=20, random_state=42), 'XGBoost': XGBClassifier(n_estimators=200, max_depth=3, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8, random_state=42, eval_metric='logloss') imported lazily}, and walk_forward_proba(X, y, make, splits=5) using TimeSeriesSplit(n_splits=5, gap=1): fit on each training block, predict_proba on the following test block, return the out-of-sample probabilities and the last model. Never fit on future data. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: walk_forward_proba returns probabilities only for the five test blocks (2020-01-06 onwards).

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Step 3: Turn probability into a position and backtest it (A4)

Ask your assistant for position_from_probability(), run() and main().

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/ml.py add position_from_probability(prob, threshold, index=None) = 1.0 where P(up) > threshold (decided at close t, so the backtester fills it at the next open), and run(symbol='SPY', threshold=0.55, models=None) for `python -m pipeline ml --threshold --models`: print samples, features, base rate and the five folds; out-of-sample accuracy, precision, AUC and share predicted up per model, with the naive 'always up' benchmark; backtest each model's positions from the first test date with common.backtest.backtest and the config cost model against buy and hold (CAGR, Sharpe, max drawdown, exposure, trades, turnover); print a VERDICT ('NO EDGE' if the best AUC < 0.53); save _outputs/Lab11_ml_probabilities.csv and chart Lab11_ml_strategy.png (equity curves and XGBoost feature importance from the last fold). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline ml --help shows --threshold and --models.

Step 4: Run the ML stage (A4, K4)

Run: python -m pipeline ml --threshold 0.55 (same as: python3 Lab11-ml-signal-classifier/ml_stage.py).

Check: 1,811 samples, 14 features, base rate 55.4%; folds test from 2020-01-06 to 2025-12-30. Out-of-sample AUC: logistic regression 0.481, random forest 0.483, XGBoost 0.497 (accuracy 0.535 / 0.542 / 0.519 versus 0.551 for always 'up').

Step 5: Read the strategy result (A4)

Read the backtest table and the VERDICT.

Check: 2020-2025, threshold 0.55: CAGR 9.19% / 7.70% / 4.00% versus buy and hold 14.98%; Sharpe 0.61 / 0.54 / 0.32 versus 0.78; turnover 83 / 75 / 96 versus 0.17; 'VERDICT: NO EDGE - keep the ML signal out of the live strategy'.

Step 6: Test the threshold - and notice the snooping risk (A4, K6)

Run: python -m pipeline ml --threshold 0.5 --models 'Logistic regression', then --threshold 0.6.

Check: Threshold 0.50: CAGR 16.82%, Sharpe 0.93, exposure 87%, 128 trades - better than buy and hold; threshold 0.60: 7.67%, Sharpe 0.60, exposure 23%, 205 trades. Choosing 0.50 AFTER seeing this table is data snooping: it must be re-tested out of sample and counted as an extra trial.

Step 7: Prove that leakage is detectable (K6)

Ask your assistant for a throw-away leakage experiment (do not save it into pipeline/ml.py).

AI prompt - paste into your AI coding assistant
In a Python shell started in labs, import pipeline.ml, wrap build_features so it also adds a feature 'leak' = tomorrow's return (bars['close'].pct_change().shift(-1)), run walk_forward_proba with the 'Logistic regression' model and print the out-of-sample AUC. Explain why the result proves the feature leaks the future, then discard the code.

Check: The leaky model scores AUC 0.997 - far too good; the honest model scores 0.481. Accuracy above about 60% on daily direction is a leakage alarm.

Step 8: Inspect the feature importance (K4)

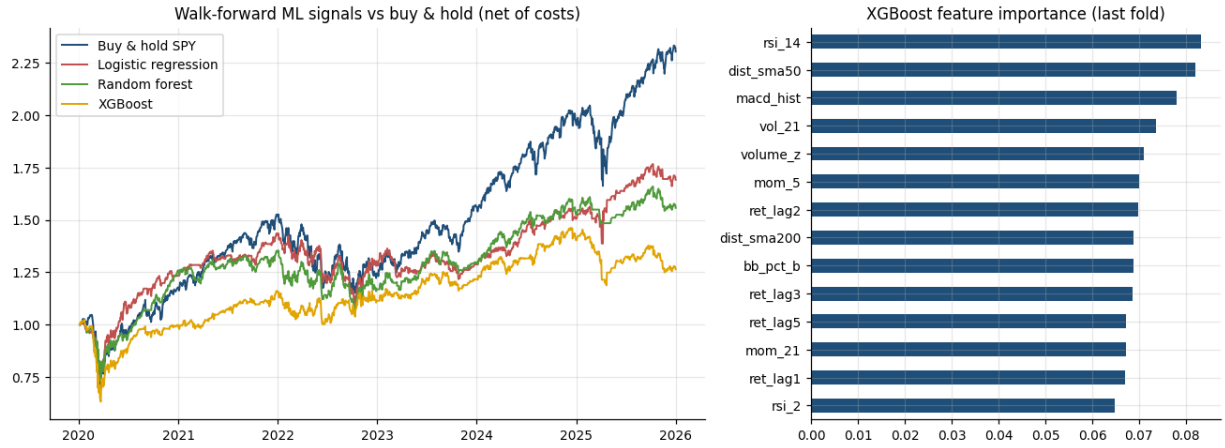
Open _outputs/Lab11_ml_strategy.png. Which features does XGBoost rely on, and does that make economic sense?

Check: The importance bars are spread across many features with no dominant one - consistent with no reliable edge.

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Expected Output (Full Code)

python -m pipeline ml --threshold 0.55: 1,811 samples, 14 features, base rate 55.4%; out-of-sample AUC 0.481 (logistic regression), 0.483 (random forest), 0.497 (XGBoost); strategies 2020-2025: CAGR 9.19% / 7.70% / 4.00% versus buy and hold 14.98%, Sharpe 0.61 / 0.54 / 0.32 versus 0.78, turnover 75-96 versus 0.17; VERDICT: NO EDGE; a leaky feature pushes AUC to 0.997; files _outputs/Lab11_ml_probabilities.csv and chart _outputs/Lab11_ml_strategy.png.



Lab 11 output (labs_fullcode/_outputs/Lab11_ml_strategy.png), offline mode on the bundled data.

Debrief

Supervised models (K4) can be trained and validated correctly (K6) and still find no edge - an honest negative result is valuable evidence (A4). Walk-forward validation stops the model learning from the future, a leakage test proves the check works, and the cost-aware backtest shows what an AUC of 0.5 is worth in money.

Troubleshooting

Symptom	What to do
xgboost not installed	Run pip install xgboost, or use --models 'Logistic regression' 'Random forest'.
Accuracy above 60%	Suspect leakage: no feature may use shift(-1), and TimeSeriesSplit needs gap=1.
Results differ in the third decimal	Random forest and XGBoost use random_state=42; library versions can still differ slightly.

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